

changed. Any improvement or decline in motor skills, speech or self-care since the last time your loved one was in therapy may mean eligibility changes in services.

- **Monitor changes in attitude and behavior.** Evaluate whether your loved one is having a hard time controlling emotions. Consult a healthcare provider to develop a plan of action.
- **Stop depression before it hinders recovery.** Post-stroke depression is common, with as many as 30-50 percent of stroke survivors depressed in the early or later phases of post-stroke. Post-stroke depression can significantly affect your loved one's recovery and rehabilitation. Consult a healthcare provider to develop a plan of action.

DIABETES

Diabetes is a chronic (long-lasting) health condition that affects how your body turns food into energy. Your body breaks down most of the food you eat into sugar (glucose) and releases it into your bloodstream. When your blood sugar goes up, it signals your pancreas to release insulin. Insulin acts like a key to let the blood sugar into your body's cells for use as energy.

With diabetes, your body doesn't make enough insulin or can't use it as well as it should. When there isn't enough insulin or cells stop responding to insulin, too much blood sugar stays in your bloodstream. Over time, that can cause serious health problems, such as heart disease, vision loss, and kidney disease.

There isn't a cure yet for diabetes, but losing weight, eating healthy food, and being active can really help. Other things you can do to help:

- Take medicine as prescribed.
- Get diabetes self-management education and support.
- Make and keep health care appointments.

Types of Diabetes

There are three main types of diabetes: type 1, type 2, and gestational diabetes (diabetes while pregnant).

Type 1 Diabetes

Type 1 diabetes is thought to be caused by an autoimmune reaction (the body attacks itself by mistake). This reaction stops your body from making insulin. Approximately 5-10% of the people who have diabetes have type 1. Symptoms of type 1 diabetes often develop quickly. It's usually diagnosed in children, teens, and young adults. If you have type 1 diabetes, you'll need to take insulin every day to survive. Currently, no one knows how to prevent type 1 diabetes.

Type 2 Diabetes

With type 2 diabetes, your body doesn't use insulin well and can't keep blood sugar at normal levels. About 90-95% of people with diabetes have type 2. It develops over many years and is usually diagnosed in adults (but more and more in children, teens, and young adults). You may not notice any symptoms, so it's important to get your blood sugar tested if you're at risk.

Gestational Diabetes

Gestational diabetes develops in pregnant women who have never had diabetes. If you have gestational diabetes, your baby could be at higher risk for health problems. Gestational diabetes usually goes away after your baby is born. However, it increases your risk for type 2 diabetes later in life. Your baby is more likely to have obesity as a child or teen and develop type 2 diabetes later in life.

Prediabetes

Prediabetes is high glucose levels but not high enough to be diagnosed with diabetes. It is estimated that one out of three adults have prediabetes and of those 15 to 30% will develop type II diabetes within five years. Additionally, prediabetes can lead to heart disease, stroke, and type II diabetes. Often with lifestyle changes, this can be reversed. Risk factors for prediabetes include being overweight, over 45 years of age, having a parent or sibling with type II diabetes, exercising less than three times per week, or having a history of gestational diabetes. To prevent or delay type II diabetes the CDC suggests losing 5 to 7% of your body weight and getting at least 150 minutes of exercise each week.

Taking care of a patient with Diabetes.

1. Keep your blood pressure and cholesterol under control.

- Like diabetes, high blood pressure can damage your blood vessels. High cholesterol is a concern, too, since the resulting damage is often worse and more rapid when you have diabetes. When these conditions team up, they can lead to a heart attack, stroke or other life-threatening conditions.
- Eating a healthy, reduced-fat and low salt diet, avoiding excess alcohol, and exercising regularly can go a long way toward controlling high blood pressure and cholesterol. Your health care provider may also recommend taking prescription medication, if necessary.

2. Schedule regular physicals and eye exams.

Schedule two to four diabetes checkups a year, in addition to yearly physical and routine eye exams. During the physical, the health care provider will ask about nutrition and activity level and look for any diabetes-related complications — including signs of kidney damage, nerve damage and heart disease — as well as screen for other medical problems. He or she will also examine feet for any issues that may need treatment. The eye care specialist will check for signs of retinal damage, cataracts and glaucoma.

3. Keep vaccines up to date.

Diabetes increases risk of getting certain illnesses. Routine vaccines can help prevent them. Ask the health care provider about:

- **Flu vaccine.** A yearly flu vaccine can help the patient stay healthy during flu season as well as prevent serious complications from the flu.
- **Pneumonia vaccine.** Sometimes the pneumonia vaccine requires only one shot. If the patient has diabetes complications or they're 65 or older, they may need a booster shot.
- **Hepatitis B vaccine.** The hepatitis B vaccine is recommended for adults with diabetes who haven't previously received the vaccine and are younger than 60. If they're 60 or older and have never received the hepatitis B vaccine, talk to the health care provider about whether it's right for them.

- **Other vaccines.** Stay up to date with the tetanus shot (usually given every 10 years). The doctor may recommend other vaccines as well.

4. **Take care of teeth.**

Diabetes may leave the patient prone to gum infections. Brush their teeth at least twice a day with a fluoride toothpaste, floss their teeth once a day and schedule dental exams at least twice a year. Call the dentist if gums bleed or look red or swollen.

5. **Pay attention to feet.**

High blood sugar can reduce blood flow and damage the nerves in feet. Left untreated, cuts and blisters can lead to serious infections. Diabetes can lead to pain, tingling or loss of sensation in the feet.

To prevent foot problems:

- Wash the patient's feet daily in lukewarm water. Avoid soaking the feet, as this can lead to dry skin.
- Dry the feet gently, especially between the toes.
- Moisturize the feet and ankles with lotion or petroleum jelly. Do not put oils or creams between the toes — the extra moisture can lead to infection.
- Check feet daily for calluses, blisters, sores, redness or swelling.
- Consult the doctor if the patient has sore or other foot problem that doesn't start to heal within a few days. If they have a foot ulcer — an open sore — see the doctor right away.
- Don't let the patient go barefoot, indoors or outdoors.

6. **Healthy diet**

Eating a healthy diet and increasing the level of physical activity should be the first steps in the management of newly diagnosed people with diabetes, and have to be maintained.

Good dietary practices in the management of diabetes mellitus:

- Appropriate diet and physical activity is important in maintaining an ideal body weight.

- Reduced intake of animal fat, salt, and refined foods is recommended.
- Locally available foods high in fiber, vegetables and fruits should be encouraged in the community.
- Eat a variety of foods in at-least three meals a day.
- Alcohol and tobacco products use should be avoided.
- Water is essential for normal body functioning. At - least eight glasses of water should be taken per day.

7. **Physical activity**

- Regular physical activity helps in;
 - Increasing insulin sensitivity thus improving blood sugar control.
 - Controlling blood pressure.
 - Improving blood flow in the heart and vessels.
 - Weight loss and maintenance of healthy body weight.
 - Giving one a sense of well-being.
- Physical activity should be regular (-3 days/week), lasting at least 20-30 min. per session, and be of at least moderate intensity.
- Activities like walking, climbing steps (instead of taking lifts) should be encouraged.
- People with diabetes require proper medical assessment before engaging in vigorous physical activity.
- Proper shoes must always be worn during exercises to avoid injury to the feet.

8. **Management of diabetes using drugs**

Diabetes is managed by use of medicines that can either be:

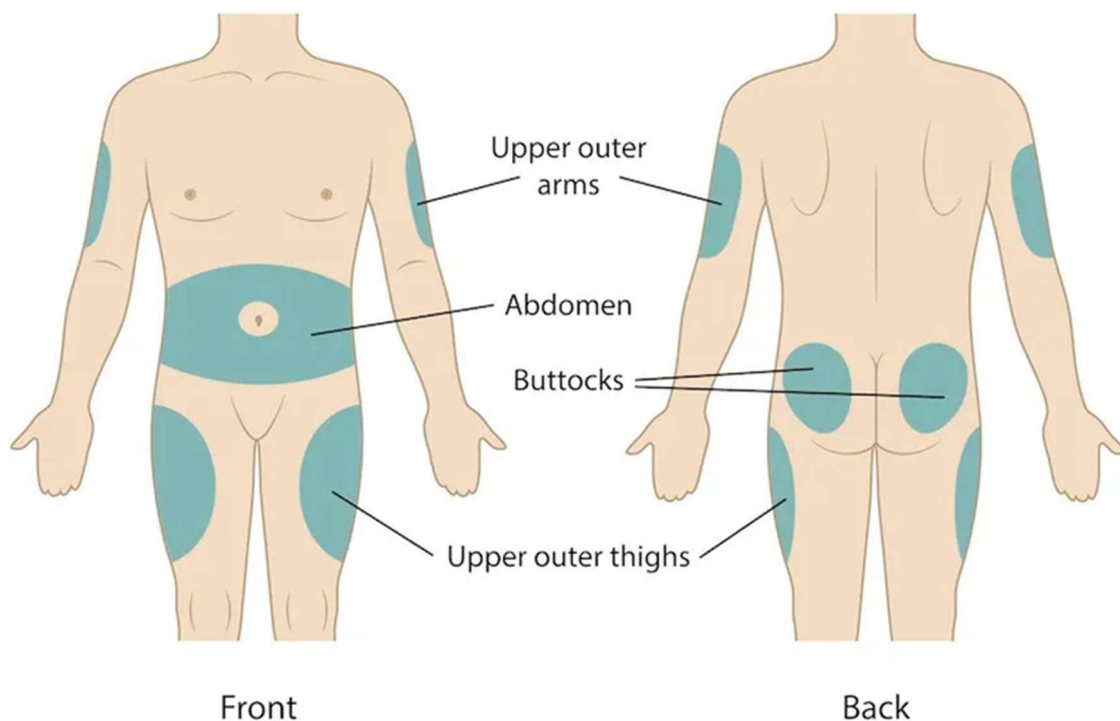
- **Oral blood sugar lowering tablets**
- **Insulin injection**

It is important to note that;

- Oral Tablets are used when an individuals' blood sugar control is not achieved using the combination of dietary modifications and physical activity/exercise.
- Patients currently using tablets to control their blood sugar may at one time in the future need insulin as these tablets may no longer work in them.
- In some cases, oral tablets or insulin injection may be given at the time of diagnosis of diabetes if the blood sugar is extremely high.
- Individuals with type 1 diabetes must be on insulin for survival.

Principles of insulin administration

Insulin injection sites



- The injection sites most commonly used for insulin are the abdomen, the thigh, and the outer upper arm.
- The rate of absorption is fastest on the abdomen.

- The thigh should be avoided as an injection site when exercise is going to be done, as this will increase the rate of absorption of the insulin.
- One area should be used for an injection at a particular time of day, e.g. the abdomen is the site for the morning injection, and the thigh is the site for the evening injection.
- Within these areas, the injection site is to be rotated. To avoid formation of a hard scar (seen as a lump).
- Injection on these scars should be avoided as insulin absorption is not predictable.
- The injection should be done at a 90-degree angle with the skin using the recommended insulin administration syringe and needle.
- The needle should be held in place for 10 seconds after injection before withdrawal.

